Database Management Systems & SQL Normalization (BCA-302 / MCA)

Uploaded By: Swaminarayan Computer Science Department | Official Course Curriculum | Peer Verified & 4.9

Unit 1: Relational Model & Relational Algebra Operators

• Selection (<2“ Filters rows satisfying predicate (<5öFW CÒ$54R"…7GVFVçG2’’à

• Projection (< “ Selects specific columns without duplicate rows (< öæ ÖP,cgpa(Students)).

• Cartesian Product (×) and Natural Join (*): Combining relational schemas on matching attributes.

• Set Operations: Union ("*), Set Difference (-), Intersection (")).

Unit 2: Functional Dependencies & Database Normalization

• 1st Normal Form (1NF): Each column must contain atomic (indivisible) values with no repeating groups.

• 2nd Normal Form (2NF): Must be in 1NF and have NO Partial Dependencies (non-prime attributes fully dependent on candidate key).

• 3rd Normal Form (3NF): Must be in 2NF and have NO Transitive Dependencies (X -> Y where Y is non-prime and X not superkey).

• Boyce-Codd Normal Form (BCNF): For every functional dependency X -> Y, X MUST be a Super Key.

Unit 3: Transaction Processing & ACID Properties

• Atomicity (All or Nothing), Consistency (State preservation), Isolation (Concurrent transparency), Durability (Persistence).

• Two-Phase Locking (2PL): Growing phase (locks acquired) & Shrinking phase (locks released) ensures serializability.